



Thermographic and cognitive assessment of fatigue in vessel traffic service operators: a socio-technical approach

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Abstract

Fatigue among vessel traffic service operators (VTSOs) poses critical risks to maritime safety, requiring objective, real-time monitoring solutions. This study integrates thermographic imaging and subjective self-reports to assess fatigue and mental workload in operational environments. A total of 23 VTSOs from two Spanish Maritime Rescue Coordination Centers (MRCCs), with an average of 7.74 years of experience, were observed for 200 h. Using facial thermography, specifically nasal temperature variations, and validated psychological assessments, we analyzed the physiological and cognitive effects of shift work. Results show that night shifts significantly increase fatigue, with a 15% increase in perceived exertion and a 2 °C decrease in nasal temperature. Automated face recognition via YOLO5Face facilitated real-time thermographic analysis, improving the accuracy of fatigue monitoring. Thermographic imaging successfully correlated nasal temperature changes with cognitive workload, demonstrating its potential as a non-invasive tool for fatigue assessment. In addition, pre-task rest was inversely related to fatigue, highlighting the importance of rest management in mitigating operator fatigue. By bridging cognitive systems engineering and socio-technical perspectives, this study provides a novel framework for fatigue assessment in safety-critical environments. The results support the integration of thermographic methods into maritime traffic management, contributing to human-centered safety technologies. Future research will explore broader applications of automated thermal analysis and additional physiological fatigue markers in high-risk industries.

Keywords Thermographic imaging · Vessel traffic service operator (VTSOs) · Fatigue monitoring · Maritime human factors · Cognitive workload · Human-technology interaction

1 Introduction

The maritime transport sector, a cornerstone of global trade, handles over 80% of global goods and employs over one million seafarers worldwide (UNCTAD 2023). With a projected growth rate of 2.1% over the next five years, the sector faces increasing operational demands that exacerbate human factors challenges, particularly fatigue and mental workload. Fatigue and mental workload are often discussed together, but are distinct concepts: fatigue is a complex state involving both psychological and physiological elements, whereas mental workload is more task-oriented (Khalamala Ibrahim Salih Barzani 2022). Fatigue results from prolonged physical or mental exertion without adequate recovery (Caldwell et al. 2019), while mental workload refers to the cognitive demands of complex tasks required to achieve a certain level of performance (Mohammadian et al. 2022). Both factors have a significant impact on the performance of maritime

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professionals, especially under the high-stakes conditions of maritime operations.

Among these professionals are Vessel Traffic Service Operators (VTSOs), specialists responsible for monitoring and managing maritime traffic to ensure safe navigation. VTSOs perform critical tasks requiring sustained attention and decision making under cognitively demanding conditions and irregular schedules, making them particularly vulnerable to fatigue and mental workload. The cognitive load on VTSOs is particularly high due to the need to process multiple streams of information (ship communications, alarms, environmental factors), which directly contributes to mental fatigue, reducing their overall cognitive capacity and increasing the likelihood of errors. Research has shown that shift work and long hours are significant contributors to fatigue among VTSOs, exacerbated by their irregular schedules that disrupt sleep patterns (Dorrian et al. 2011; Yoo and Kim 2021).

Studies also suggest that high cognitive workloads combined with long working hours increase the risk of human error in maritime operations (Li et al. 2020; Crestelo Moreno et al. 2023; Goerlandt and Liu 2023). For example, a report by the Cardiff Research Programme on seafarer fatigue found that 22% of watchkeepers get less than four hours of sleep in a 24-hour period (Smith et al. 2006). In this sense, VTS operations, characterized by irregular shifts, further disrupt sleep patterns (Yoo and Kim 2021). In addition, VTSOs operate at their cognitive limits, processing vast amounts of information, which increases stress, mental workload, and fatigue (Kum and Furusho 2014; Xia et al. 2025).

In 2022, RightShip, a maritime risk management and environmental assessment organization, recorded 2400 maritime incidents with known locations, half occurring within port and terminal boundaries, including 813 on docked vessels. Ship collisions accounted for 31% of incidents in ports globally and 37% in Mediterranean ports (Rightship 2022). Similar trends can be seen in South Korea, where high workloads contribute to maritime casualties. The Korea Maritime Safety Tribunal (KMST) reported 216 ship collisions in South Korean ports between 2016 and 2020, with 19.7% occurring in VTS areas (Yoo and Kim 2021). These incidents underscore the critical role of VTSOs in ensuring safe navigation and highlight the need for continuous monitoring and effective fatigue management strategies (Du et al. 2020; Adland et al. 2021; Zhang et al. 2023). Traditional fatigue assessment methods, such as self-report questionnaires, remain widely used but have notable limitations due to their subjectivity and reliance on retrospective reporting, particularly in dynamic operational environments such as maritime traffic management (Li et al. 2020; Adão Martins et al. 2021). While subjective measures provide some insight into fatigue levels, they fail to capture real-time, objective data. To fill this gap, researchers have turned to

neurophysiological measures such as electroencephalography (EEG) and heart rate variability (HRV), as well as wearable biosensors and eye-tracking technologies, to provide continuous, real-time monitoring of fatigue and workload (Xu et al. 2018; Kundinger et al. 2020; Balam et al. 2021; Giorgi et al. 2023). Although these methods provide valuable insights into physiological fatigue markers, some can be intrusive, requiring physical contact with the operator or causing discomfort over prolonged use. In response, non-invasive approaches, such as thermal imaging, offer a promising alternative. Thermal imaging uses infrared radiation emitted from the skin to detect fatigue-related physiological changes that correlate with stress, cognitive workload and fatigue levels (Lahiri et al. 2012; Villa et al. 2020; Kesztyüs et al. 2022). By capturing real-time variations in facial temperature, thermography provides an innovative, non-intrusive method for monitoring fatigue in high-stakes environments such as maritime traffic management (Veltman and Vos 2005).

This study integrates thermographic analysis with subjective self-reports to assess fatigue in VTSOs under real-world operational conditions. By examining key objective parameters such as facial temperatures and nasal thermal fluctuations, and utilizing advanced computational tools such as the YOLO5Face algorithm for thermal data analysis, this research provides a novel approach to fatigue assessment. The results contribute to measurement science by demonstrating the utility of thermography as an efficient, non-invasive tool for capturing real-time physiological responses to mental workload and fatigue, which could revolutionize fatigue management practices. In addition, the study provides evidence-based recommendations for optimizing workload management and staffing strategies, ultimately improving maritime safety and performance. These findings also have broader implications for high-risk sectors that rely on human oversight of complex systems, highlighting the transformative potential of innovative, measurement-based approaches to fatigue management.

2 Thermography as a valid tool for scientific studies

Thermography is a technique used to measure temperatures at a distance without physical contact. It captures infrared radiation (electromagnetic and thermal radiation) using specialized thermal cameras. A key advantage of thermography is its non-invasive nature, which improves subject comfort by eliminating physical contact. In addition, thermal imaging produces less signal noise than other fatigue assessment techniques, such as electroencephalogram (EEG) and electrooculography (EOG), where signal processing and filtering

are critical (Yaseen et al. 2013; Çınar 2021; Salvi et al. 2024). Unlike EEG and EOG, which measure bioelectrical activity and are prone to physiological artifacts and electrical noise, thermography captures infrared radiation, making it less susceptible to external electromagnetic interference. While EEG/EOG signals often require extensive pre-processing (e.g., filtering, artifact rejection), thermographic data primarily requires environmental standardization to ensure accuracy (Mazdeyasna et al. 2023).

Thermography gained significant attention after World War II, initially for military applications such as improving vision in low-light conditions through electronic sensors (Matchar 2017). It quickly transitioned into medicine, with its first use in the mid-20th century for non-invasive measurement of body temperature (Boas 1964). Since then, thermography has expanded into various fields, including construction, industrial materials testing, and even commercial application; improving the quality of the construction of new buildings; in commercial and industrial areas: study of materials (Reynolds 1986; Titman 2001; Maldague 2012; Taylor et al. 2013).

Recent advances in the miniaturization and quality of thermal cameras now allow studies to be conducted during normal activity without interruption. Thermography is increasingly being used in medical fields such as cancer detection (Rastghalam and Pourghassem 2013), ophthalmology (Tan et al. 2009), management of conditions like complex regional pain syndrome (CRPS) (Gulvich et al. 2004), Raynaud's phenomenon (Merla et al. 2002; Papaléo et al. 2016), rheumatoid arthritis (Boas 1964; Borojevic et al. 2011) and diabetic foot syndrome (Netten et al. 2013). In cardiovascular health, thermography also shows promise in predicting disease (Thiruvengadam et al. 2014).

Aligned with the goals of this study, thermography has proven valuable for detecting somatic markers in emotional and cognitive task performance studies (Khan et al. 2006; Jukiewicz et al. 2021). Research has focused on using thermographic data to assess mental workload in the orofacial region, highlighting key points of interest (POIs) that can be indicative of cognitive stress or fatigue (Salazar-López et al. 2015; Gomez and Moliné Segovia 2018; Aryal and Becerik-Gerber 2019) (Fig. 1). Facial skin temperature, which is regulated by the autonomic nervous system, provides a physiological indicator of changes in blood flow and can be measured remotely using infrared thermography (Masaki et al. 2021; Tian et al. 2023). In particular, the anatomy of the face, including the epidermis, dermis, and subcutaneous tissue, contributes to unique facial thermal patterns that correspond to different activities (Pavlidis and Levine 2002; Stanić and Geršak 2025).

For example, temperature variations in the forehead region correlate with stress levels (Puri et al. 2005), while variations in the nasal region are associated with cognitive

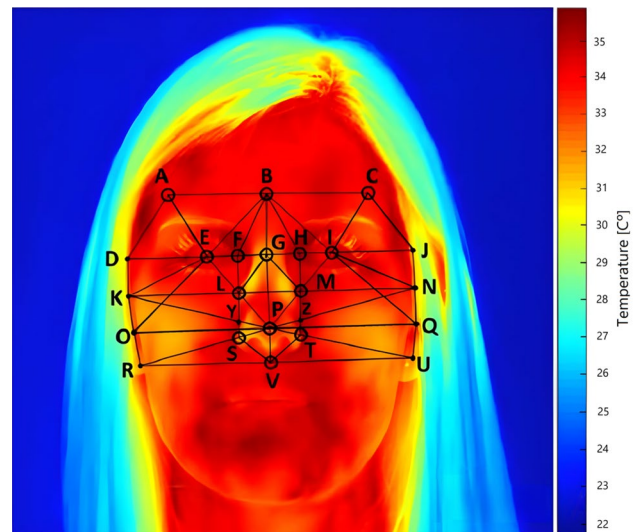


Fig. 1 Points of interest (POI) in thermographic images in the orofacial zone (Source: Marinescu et al. 2018)

and emotional responses, including fatigue (Murai et al. 2015; Marinescu et al. 2018; Diaz-Piedra et al. 2019).

Based on these findings and the thermographic protocols outlined below, this study develops a methodology to detect temperature changes associated with fatigue by focusing on these POIs.

2.1 Thermographic protocols

Accurate and reliable thermographic measurements are critical to the advancement of fatigue monitoring. To ensure accuracy, this study follows strict protocols based on established guidelines such as the International Bureau of Weights and Measures (BIPM) *Best Practice Guide for Thermal Imagers* and recommendations from previous studies on thermographic image acquisition and analysis, including those established by the University of Glamorgan (Ammer 2008; Fernández-Cuevas et al. 2015; Korman et al. 2016; Moreira et al. 2017; Pusnik et al. 2023). These protocols address both environmental and human factors to ensure standardized conditions and minimize measurement errors.

2.1.1 Calibration of the thermal camera and algorithm

The thermal camera used in this study was regularly calibrated by a certified laboratory to ensure its accuracy and optimal performance, adhering to standard calibration protocols for thermal imaging systems. This ensures that the captured thermal data remains reliable and accurate for the algorithm's facial landmark detection. Importantly, the algorithm used for facial landmark detection does not require

manual calibration before each session. It automatically detects key facial landmarks, including the left and right eye, nasal region, and forehead, using a pre-trained YOLO5Face model. This feature enhances its applicability for real-time assessments, making it feasible for operational environments where pre-task calibration may not be practical.

2.1.2 Environmental controls

Key environmental factors were controlled to reduce noise in the thermal data. Room temperature was maintained between 18°C and 25°C, with relative humidity between 45% and 60%. While the influence of relative humidity is often overlooked in human infrared thermography (IRT) studies (Uematsu et al. 1988; Atmaca and Yigit 2006), it was considered here as a potential factor affecting thermal readings.

Additionally, a minimum distance of two meters was maintained between the camera, subject, and image processing equipment to prevent heat accumulation and distortion of the thermal readings.

2.1.3 Emissivity calibration

Thermal cameras do not directly measure the temperature of the target surface (e.g., human skin), but instead capture the total radiation within their field of view (FOV). This radiation includes emissions from the target surface, reflected radiation from the environment, and radiation absorbed and emitted by the atmosphere. According to Kirchhoff's Law, the atmosphere absorbs a portion of the radiation and emits its own radiation, which must be accounted for in the measurements (Piccinelli et al. 2024). The total radiation detected by the camera, denoted as W , is represented by the following equation:

$$W = \epsilon \tau W_{obj} + (1 - \epsilon) \tau W_{amb} + (1 - \tau) W_{atm} \quad (1)$$

where

- ϵ represents the emissivity of the target surface (i.e., human skin),
- τ is the transmission through the atmosphere,
- W_{obj} refers to the radiation emitted by the target object (skin),
- $(1 - \epsilon)$ corresponds to the reflectivity of the target surface (the portion of radiation reflected from the surface),
- W_{amb} is the radiation emitted by the surroundings (ambient radiation),
- W_{atm} refers to the radiation emitted by the atmosphere.

Human skin is highly efficient at absorbing infrared radiation and behaves similarly to a black body in this range, with an emissivity close to 1 (Mitchell et al. 1967). This high emissivity is likely due to the skin's rough texture and its high-water content (Mansi et al. 2021). The commonly accepted emissivity value for clean, dry skin is 0.98, although some studies use values slightly closer to 1 depending on the conditions (Charlton et al. 2020).

2.1.4 Personal recommendations and subject control

To minimize external influences on skin temperature, participants were asked to follow personal guidelines. These included wearing hair ties to keep hair away from the forehead and neck; avoiding skin products such as creams, lotions, or deodorants; and avoiding physical exertion, hot or cold beverages, caffeine, or alcohol prior to the study. Smoking was also prohibited due to its potential impact on thermal measurements. All of these factors can affect body temperature and therefore the accuracy of thermographic readings (Ammer et al. 2003; Gomes Moreira et al. 2017; Faria et al. 2021).

While these protocols aim to standardize measurement conditions, it is critical to address the uncertainty in thermographic measurements. Uncertainty arises from factors such as environmental variations, emissivity calibration, and subject-specific characteristics, all of which can contribute to slight variations in thermal measurements. Incorporating uncertainty analysis, as described in the Methodology section, strengthens the reliability of thermographic data and helps to contextualize its results.

3 Materials

This study employed several materials to assess operator fatigue, focusing on both physiological and subjective measures. These included a thermal imaging camera, a set of standardized self-report questionnaires, and a custom-developed Integrated Workload Scale (IWS), designed specifically for this study in collaboration with two experienced Vessel Traffic Service Operators (VTSOs) with over 20 years of operational experience.

3.1 Thermal imaging camera

The thermal imaging camera used in this study was the FLIR E60, which offers a resolution of 320 x 240 pixels at 60 Hz and is sensitive enough to detect subtle temperature changes, even in low-contrast environments. This capability is essential for assessing physiological responses under different conditions. Thermal image sequences were recorded using

Table 1 FLIR E60 Technical Data (Source: www.flir.com)

Parameter	Measurement
IR Resolution	320×240 (76.800 pixels)
Thermal Sensitivity	<0.05 °C @ +30 °C (+86°F) / 50 mK
Temperature Range	−20 °C to +650 °C (−4°F to +1202°F)
Accuracy	±2 °C (±3.6°F) or ±2%
Field of view / focal length	25°×19° / 0.4 m (1.31 ft.)
Video Camera (no backlit)	3.1 MP

the dedicated FLIR ThermoCAM Researcher 2.10 software, and analyses were performed using YOLO5Face (Qi et al. 2021), a Python-based algorithm specifically designed for this purpose. The technical specifications of the FLIR E60 camera are depicted in the table below (Table 1).

3.2 Questionnaires

To complement the thermographic measurements, several standardized self-report questionnaires were used to assess the subjective experiences of the participants. These validated instruments focus on various aspects of fatigue, workload, emotional state, and circadian rhythm preferences. The following questionnaires were used:

Stanford Sleepiness Scale (SSS): This scale assesses a participant's current level of sleepiness at the time of administration. Developed by Hoddes et al. 1973, the SSS consists of a single-item seven-point Likert scale (1–7) where respondents indicate their level of alertness or drowsiness. The scale ranges from 1 (feeling active, vital, alert, or wide awake) to 7 (almost in reverie, sleep onset soon, lost struggle to remain awake). The SSS is widely used in sleep research and clinical settings to measure momentary fluctuations in sleepiness, making it suitable for repeated measurements throughout an experiment. Higher scores indicate greater sleepiness, and values of 4 or above suggest potential sleep deprivation.

Epworth Sleepiness Scale (ESS): The ESS measures general daytime sleepiness by assessing an individual's likelihood of dozing off in different situations (Johns 1991). The scale consists of eight items, each rated on a 4-point scale (0–3), with the total score ranging from 0 to 24. A higher score indicates increased daytime sleepiness. The interpretation of the scores follows a structured progression from normal to excessive sleepiness: Lower Normal (0–5), Higher Normal (6–10), Mild Excessive (11–12), Moderate Excessive (13–15), and Severe Excessive (16–24).

NASA Task Load Index (NASA-TLX): A widely accepted tool for assessing perceived workload across multiple dimensions, including mental, physical, and temporal demands, performance, effort, and frustration (Noyes and

Bruneau 2007; Bradley and Lang 1994; Ruiz-Rabelo et al. 2015).

The NASA-TLX has been frequently used as a standard measure in workload assessment due to its comprehensive evaluation of task-related demands. The interpretation of scores follows a continuous scale from 0 to 100, linking increasing workload demands with higher values: Low (0–9), Medium (10–29), Somewhat High (30–49), High (50–79), and Very High (80–100).

Self-Assessment Manikin Scale (SAM): The SAM is a non-verbal pictorial assessment technique designed to measure the pleasure, arousal, and dominance dimensions of emotional responses (Bradley and Lang 1994). In this study, only pleasure and arousal dimensions were used because SAM ratings can be used to directly plot any object or event in a two-dimensional “affective space” (Lang et al. 1993). The SAM uses a five-point scale where participants indicate their emotional state. For pleasure, the scale ranges from 1 (unpleasant) to 5 (pleasant), while for arousal, it ranges from 1 (calm) to 5 (excited).

Borg Rating of Perceived Exertion (Borg RPE): This scale measures perceived exertion during physical activity, ranging from 6 to 20, with values correlating with heart rate (Borg 1982; Williams 2017). The RPE scale is commonly used in occupational health and sports medicine research, as it allows individuals to rate their physical effort subjectively. The interpretation of scores follows a structured scale, linking exertion levels to physical intensity: No exertion (6), Extremely Light (7–8), Very Light (9–10), Light (11–12), Somewhat Hard (13–14), Hard (15–16), Very Hard (17–18), and Maximal Exertion (19–20) (Table 2).

These self-report tools are widely used in fatigue and workload (Stemberger et al. 2010; Di Stasi et al. 2020; Behrens et al. 2023). With the exception of the ESS, which assesses baseline daytime sleepiness, all questionnaires were administered before and after each observation period, as detailed in the *Procedures* section, to evaluate changes in participants' subjective experiences. A comprehensive description of all tests used in this study can be found in *Appendix A*.

3.3 Integrated workload scale (IWS)

The Integrated Workload Scale (IWS) was developed specifically for this study to capture the complexity of Vessel Traffic Service Operator (VTSO) tasks. Designed to assess the challenges faced by VTSOs in their operational environment, the IWS evaluates cognitive demands, communication, coordination, and decision making under both routine and high-stress conditions. While similar to other established workload tools such as the NASA Task Load Index (NASA TLX), the IWS is tailored to the unique operational needs of VTSOs.

Table 2 Summary of Applied Scales

Scale	Description	Administration	Range & Interpretation
Epworth Sleepiness Scale (ESS) (Johns 1991)	Assesses general daytime sleepiness through 8 situational likelihood ratings	Completed at baseline to establish participants' predisposition to daytime sleepiness	0–24 scale; 0–5 (lower normal), 6–10 (higher normal), 11–12 (mild), 13–15 (moderate), 16–24 (severe excessive sleepiness)
Stanford Sleepiness Scale (SSS) (Hoddes et al. 1973)	Measures momentary sleepiness on a 7-point Likert scale. Used in sleep research to assess alertness fluctuations	Administered before and after each watch period to track alertness changes	1 (alert) to 7 (extreme sleepiness); scores ≥ 4 indicate sleep deprivation
Self-Assessment Manikin (SAM) (Bradley & Lang 1994)	Non-verbal pictorial scale measuring pleasure and arousal dimensions of emotional response	Administered before and after each watch period to assess emotional state changes	Pleasure: 1 (unpleasant) to 5 (pleasant); Arousal: 1 (calm) to 5 (excited)
NASA Task Load Index (NASA-TLX) (Hart & Staveland, 1988)	Measures perceived workload across six dimensions: mental, physical, temporal demand, performance, effort, and frustration	Administered before and after each watch period to evaluate task-related workload	0–100 scale; Low (0–9), Medium (10–29), Somewhat High (30–49), High (50–79), Very High (80–100)
Borg Rating of Perceived Exertion (RPE) (Borg 1982)	Assesses perceived physical exertion during tasks, correlating with heart rate	Administered before and after each watch period to evaluate subjective physical effort	6 (no exertion) to 20 (maximal exertion); Increasing values indicate greater perceived effort

To ensure the validity of the scale, the Thurstone Equal Interval Method was used in its development (Thurstone 1927). This method was applied with the direct involvement of two experts with extensive experience in VTSO operations. This method, used with the direct input of two VTSO experts, allowed the scale to be structured to reflect an equal level of perceived workload across task categories. The experts' feedback ensured that the scale accurately represented VTSO work demands. In addition, the design of the IWS was informed by similar workload measurement studies in specific operational contexts (Pickup et al. 2005). The IWS was tested under real-world conditions prior to this study, where it was validated by comparing temperature values and self-reported mental and physical fatigue against workload levels. This process ensured that the scale accurately reflects the complexity of VTSO tasks in both routine operations and high-stress situations.

The IWS assesses both cognitive demands, such as monitoring multiple vessels, and the impact of disruptive events, such as maintenance or emergencies, on workload. This multidimensional approach allows the scale to capture the nuances of VTSO tasks and provide a clearer understanding of workload variations during routine and high-stress conditions. The operational tasks for VTSOs related to the workload are outlined in the table below (Table 3), which summarizes the various categories of tasks VTSOs perform and their associated workload levels.

The development of the IWS was based on the structure of the NASA TLX, in particular its six dimensions: mental, physical, and temporal demands, as well as performance, effort, and frustration. This comprehensive framework was adapted to the VTSO context to provide a more accurate assessment of workload in both normal operations and emergency events. By integrating expert feedback and operational data, the IWS accurately reflects the complexity of VTSO responsibilities across multiple domains. This ensures that the scale provides a nuanced assessment of workload variation, providing a comprehensive understanding of workload intensity in both normal and emergency situations.

4 Methods

4.1 Participants

This study involved 23 Vessel Traffic Service Operators (VTSOs) from two Spanish Maritime Rescue Coordination Centers (MRCCs) located in Cartagena and Gijón, representing 7.67% of the national VTSO population. The participants had a mean age of 42 years ($SD = 9.56$) and a mean VTS experience of 7.74 years. Gender and shift schedules were evenly distributed, with participants working 7-hour morning and afternoon shifts and 10-hour night shifts, followed by three days off.

Table 3 Communication and Coordination Tasks of Spanish VTSOs. (Own source. Illustrative images of vessels generated by artificial intelligence using DALL·E, 2025)

Task	VTSO Function	COMMS	Information required	Workload Level IWS / NASA	Visual
Fishing Vessel (F)	Ship Reporting Traffic monitoring Logbook	Native language VHF	Ship identification, Number of crew members, Arrival and departure schedule	1 / Low	
Yacht Club (Y)	Yacht and boats Report- ing Traffic monitoring Logbook	Native language VHF/Telephone	Yacht name, Number of crew members, Arrival and departure schedule	1 / Low	
Barge Services (B)	Boat Reporting Coordination Check Clearance Traffic monitoring Logbook	Native language VHF	Boat name Vessel to be serviced and Schedule expectations	1 / Low	
Cleaning Boats (C)	Boat Reporting Check Clearance Traffic monitoring Logbook	Native language VHF	Boat name Work area and time schedule	1 / Low	
Tourist Boat (T)	Boat Reporting Traffic monitoring Logbook	Native language VHF	Boat name Number of Crew mem- bers & passenger, Arrival and departure schedule	1 / Low	
Shipyard (S)	Ship Reporting Pilots Coordination Traffic monitoring Logbook	English language VHF	Ship identification, Num- ber of crew members, Departure schedule	2 / Medium	
Navy Ship (N)	Ship Reporting Pilots Coordination Clearance Traffic monitoring Logbook	English language VHF	Ship identification, Number of crew members, Arrival and departure schedule	2 / Medium	
Visitors (V)	Managing the pres- ence of visitors (e.g., maintenance personnel, inspectors), ensuring operational tasks are not disrupted	Maintaining com- munication and coordination	Visitor identification, Purpose of visit, Time in the control room, Areas affected by the visit, Adjustments to operational priorities	3/ Somewhat High	
Merchant Ships (M)	Ship Reporting Pilots Coordination Check Clearance Traffic monitoring Logbook	English language VHF	Ship identification, Num- ber of Crew members & passenger, Arrival/departure Sched- ule, Sailing Plan, Cargo onboard, etc. (see IMO A.851)	4 / High	
Emergency Operations	Emergency operations (e.g., crisis manage- ment, vessel collision, rescue coordination, incident handling)	English and/or native language, VHF	Immediate vessel status, Crew and passenger safety information, Current and forecasted weather conditions, Emergency protocols and plans	5 / Very High	

Table 4 Summarizes thermographic hours per shift at both MRCCs

Shift	MRCC Cartagena	MRCC Gijón
Morning	84 h (12 recordings)	28 h (4 recordings)
Afternoon	56 h (8 recordings)	14 h (2 recordings)
Night	50 h (5 recordings)	10 h (1 recording)

4.2 Data collection

Data were collected during over 200 hours of shift observations across morning, afternoon, and night shifts (Table 4). Measurements included:

- **Thermographic Imaging:** Real-time physiological data was collected using a FLIR E60 thermal camera, focusing on nasal temperature variations as an indicator of fatigue.
- **Self-Reported Questionnaires:** Standardized instruments, including the NASA Task Load Index (NASA-TLX) and Stanford Sleepiness Scale (SSS), assessed subjective perceptions of workload and fatigue.
- **Integrated Workload Scale (IWS):** A novel instrument developed for this study provided real-time assessments of mental workload.

4.3 Analysis

4.3.1 YOLO5Face for thermal image processing

Thermal image data were recorded using FLIR ThermoCAM Researcher 2.10 software, which provided the initial data capture and ensured the reliable acquisition of thermal images. After the recording phase, the YOLO5Face algorithm (Qi et al. 2021), a deep learning-based algorithm derived from YOLOv5 (Jocher 2020) was employed for post-analysis. This algorithm processed the recorded video frames and analyzed key data, including temperature changes and facial movements. YOLO5Face was chosen for its ability to detect facial landmarks efficiently, even in challenging environments. The model was specifically trained on the Thermal Face in the Wild (TFW) dataset (Kuzdeuov et al. 2021), which contains over 9,982 annotated thermal images, including faces captured under various conditions (e.g., occlusion, head pose variations, and different weather settings). This broad dataset enables the model to generalize more effectively to realistic scenarios, making it well-suited for analyzing the orofacial region in our study. The TFW dataset's diversity, encompassing various age groups, genders, ethnicities, and environmental conditions (e.g., ambient temperatures and lighting) further supports the model's



Fig. 2 Points of interest in the orofacial region analyzed in the study, including the nasal area (n), frontal area (fh), left eye (l_e), right eye (r_e), and background temperature (bg)

efficacy in real-world applications. However, as with all datasets, some demographic variations may be underrepresented, which is acknowledged as a potential limitation.

To adapt YOLO5Face for thermal imagery, we leveraged the model pre-trained on TFW, as it addresses the domain gap between visible light and thermal images. The model identifies key facial landmarks, including the nasal (n) and frontal (fh) points (Fig. 2), which are reliable physiological indicators of fatigue (Gomez 2008; Salazar-López et al. 2015; Marinescu et al. 2018, b; Aryal and Becerik-Gerber 2019; Ordun et al. 2020).

The nasal (n) and frontal (fh) points were chosen because of their well-documented physiological responses to cognitive and emotional stress. Studies have shown that under high mental workload, vasoconstriction occurs in the nasal region, leading to a measurable decrease in temperature (Murai et al. 2015; Marinescu et al. 2018; Diaz-Piedra et al. 2019). Conversely, the frontal (fh) point, which is associated with increased blood flow during emotional arousal, remains stable under workload-induced fatigue and serves as a control measure (Puri et al. 2005). This differential thermal response makes these POIs particularly effective for detecting fatigue.

Additionally, the left eye (l_e) and right eye (r_e) points were detected as geometric references to improve the localization accuracy of primary points of interest (POIs). The inclusion of background temperature (bg) served as an environmental control.

To assess the model's accuracy in detecting facial key points in thermal images, we referenced the evaluation metrics provided in the TFW study, where the model achieved an average precision (AP) of 97% on a test dataset of outdoor thermal images. In the present study, we conducted

additional verification by manually annotating POIs on a subset of frames. Specifically, 200 points were selected by sampling frames approximately every 15 min, and they were then compared to 8000 temperature points that were automatically detected by the algorithm (one every ~3.15 seconds). This process ensured the accuracy of the model in identifying key facial landmarks. The results showed a deviation of less than 0.1°C between the manually extracted and algorithm-detected temperature readings, confirming the model's reliability within the tolerance limits of thermographic analysis.

4.3.2 Data processing

To address fluctuations in the thermal data, this study employed a combination of real-time data capture using FLIR ThermoCAM Researcher 2.10 software and post-analysis with YOLO5Face. The FLIR software recorded the thermal images, while YOLO5Face was subsequently used to process the recorded video frames. This hybrid approach enabled efficient monitoring of fatigue over time by extracting and analyzing data such as temperature changes and facial movements.

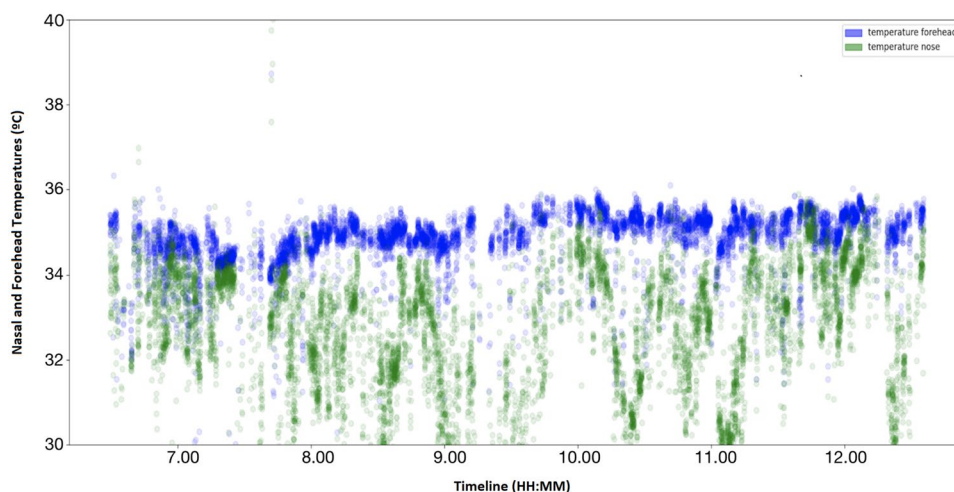
Several strategies were implemented to reduce the impact of motion artifacts, occlusions, and variations in facial positioning, thereby ensuring the reliability of the thermal data:

1. **Standardized Camera Positioning:** The thermal camera was positioned 90 cm from the subject at a consistent angle to minimize geometric distortions and ensure focus consistency, following established protocols (Ammer and Ring 2007; Priego Quesada et al. 2016).
2. **Controlled Environment:** Environmental conditions, including ambient temperature and humidity, were carefully regulated to stabilize thermal readings and minimize external influences on data quality.
3. **Subject Preparation Protocols:** Participants followed standardized pre-recording protocols, which included avoiding physical exertion, the use of certain skin products, and the consumption of caffeine or alcohol. These precautions helped reduce potential variables affecting thermal measurements.
4. **Calibration and Distance Maintenance:** The thermal camera was regularly calibrated to maintain measurement accuracy. A minimum distance from each subject was preserved throughout the recording sessions to avoid heat accumulation and thermal interference, thus supporting data integrity.

To reduce the computational load, a frame downsampling ratio has been implemented. Given the frame rate of the FLIR camera (7.5 fps), long duration recordings result in thousands of frames. Since physiological temperature changes are gradual, a high frame rate is unnecessary (Stanić and Geršak 2025). Instead, frames were sampled at predefined intervals, such as one per second (ratio ≈ 0.1) or one per minute (ratio ≈ 0.002), depending on the needs of the analysis to maintain accuracy while improving processing efficiency (Lötsch et al. 2021).

To illustrate the processing pipeline, we present a representative 6-hour data set sampled at one frame per second. As shown in Fig. 3, the raw thermal data alone are not immediately interpretable. After frame extraction, the YOLO5Face model detects facial landmarks in 8-bit grayscale images. These landmarks are then mapped to corresponding 16-bit thermal frames to obtain temperature values from the forehead, nose, and background regions. Then, to determine when another face appears in the frame, we track the motion of the detected face's left eye. The assumption is that significant variations in the left eye's position indicate the presence of another face. However, this approach has a limitation: if

Fig. 3 Evolution of the orofacial temperature over the watch—raw data



the new face is close enough to the subject's, the variation may be misinterpreted as natural head movement.

To detect such variations, we conduct a statistical analysis of eye movement by calculating position changes between consecutive frames ('dX'). We then compute the mean ('m') and standard deviation ('std'), marking any point beyond 'm + 4std' as an outlier.

These outliers, which suggest a possible face switch, are highlighted in red. However, two key challenges arise during landmark acquisition:

1. **Multiple Face Detection;** When multiple faces appear, the model may switch between them, resulting in artificial temperature "jumps". We addressed this by applying a 5-min moving average filter (2.5 min before and after each point), smoothing fluctuations while preserving long-term trends.
2. **No Face Detected;** In frames without a detectable face, landmark coordinates and temperatures are missing. Various imputation strategies were tested, including assigning zero, using background temperature, or holding the previous value. For this study, we used the zero-value approach for marking absent faces. These zero-value points were later excluded from plots, and for averaging purposes, replaced with the last known temperature.

Following these steps, the resulting temperature curves were significantly smoothed, as shown in Fig. 4, enhancing the interpretability of the thermal data across long durations.

The final processed data were stored in CSV files, containing timestamps, facial landmark coordinates, and their corresponding temperature values.

4.3.3 Statistical analysis

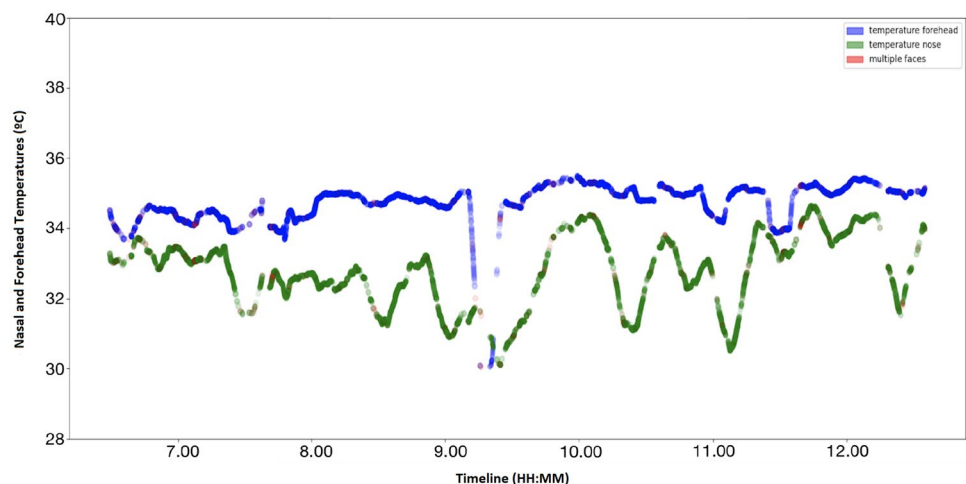
Statistical analyses were conducted in RStudio (RStudio Team 2020) using packages such as readxl (Wickham and Bryan 2023), ggpubr (Kassambara 2023a), ggplot2 (Wickham 2016), ggdist (Kay 2024), ggrain (Judd et al. 2024), corrr (Kuhn et al. 2022), ggcorrplot (Kassambara 2023b), and dplyr (Wickham et al. 2023). To visualize how the variables are related, correlation plots were performed. T-test and Cohen's d analyses were performed to test the statistical significance between variables and the effect size respectively. ANOVA tests were also performed to see the differences between variables. These analyses examined relationships between nasal temperature, shift timing, and subjective workload measures derived from self-reported questionnaires, providing insights into the interplay between physiological and perceived fatigue indicators.

4.4 Procedure

Thermographic data were collected according to established protocols for human studies, as previously cited. The camera setup, positioned 90 cm from the subject at a fixed angle, ensured consistent focus and minimized geometric distortion (Fig. 5). An emissivity coefficient of 0.98 was used for clean, dry facial skin, following values reported in the literature (Mitchell et al. 1967; Charlton et al. 2020; Mansi et al. 2021). This value is widely used in thermographic studies due to the high infrared absorption of human skin and its similarity to a black body (Bernard et al. 2013; Stanić and Geršak 2025). Adjustments were made for variations in different regions of the face to account for subtle temperature differences.

At the beginning of their shifts, participants completed demographic and self-report questionnaires following a 15-minute acclimatization period. During this period,

Fig. 4 Evolution of the orofacial temperature over the watch using a moving average of 300 s



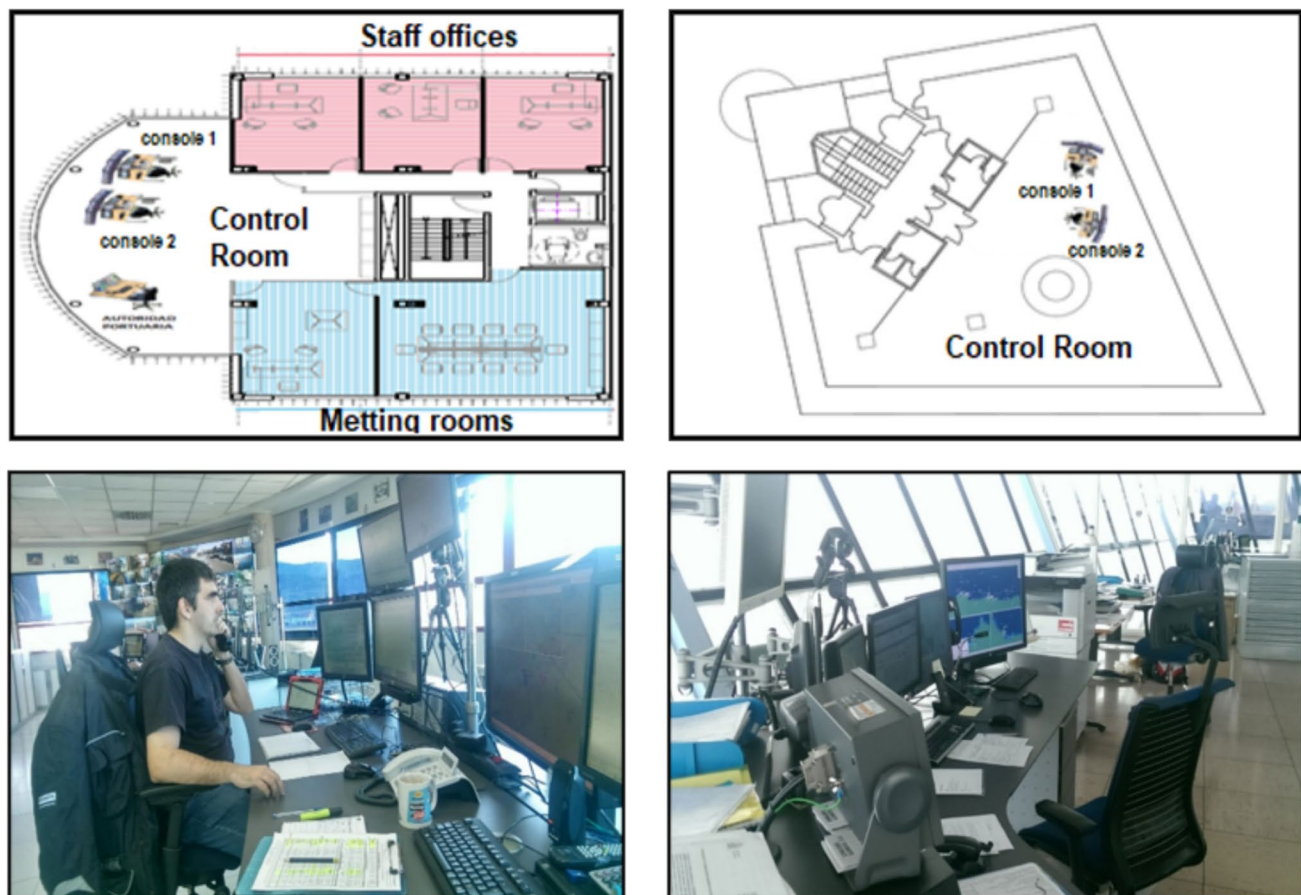


Fig. 5 Control room plans and console images of MRCC Cartagena (left) & Gijón (right)

the participants remained in a controlled environment to allow their physiological responses to stabilize prior to the commencement of data collection (IACT 2002). These pre-shift assessments established baseline data for perceived exertion and other psychological factors. Similarly, post-shift assessments were conducted immediately after participants completed their shifts to ensure an accurate evaluation of changes in perceived exertion and fatigue. Conducting assessments without delay minimized potential variability due to external factors and provided a direct comparison to pre-shift baseline values. The principal investigator remained present throughout the operators' shifts, recording all incidents that occurred and overseeing the administration of pre- and post-shift assessments. This ensured consistent and reliable data collection and immediate documentation of any relevant events that could impact fatigue levels. Continuous thermographic monitoring was performed throughout the shift to track fluctuations in fatigue levels. Additionally, observations of ship-to-shore communications contributed to the real-time assessment of IWS. The study protocol was approved by the Scientific and Ethical Committee

of the Universidad Politécnica de Cartagena, ensuring that participants had the right to volunteer and withdraw at any time.

5 Results

5.1 Subjective fatigue levels and shift impact

The present study indicated that night shifts were associated with significantly higher levels of fatigue in comparison with morning and afternoon shifts, as indicated by the Borg Rating of Perceived Exertion (RPE). A comparative analysis revealed that night shift workers demonstrated a 15% increase in their RPE scores after shifts, while those working morning shifts exhibited a 10% increase ($p < 0.01$, paired t-test). The figure below illustrates the pre- and post-shift Borg RPE scores across all shifts, highlighting the significantly higher exertion levels observed during the night shifts (Fig. 6).

The Cohen's d analysis (0.68) shows a moderate to large effect size. The ANOVA analysis found that there were

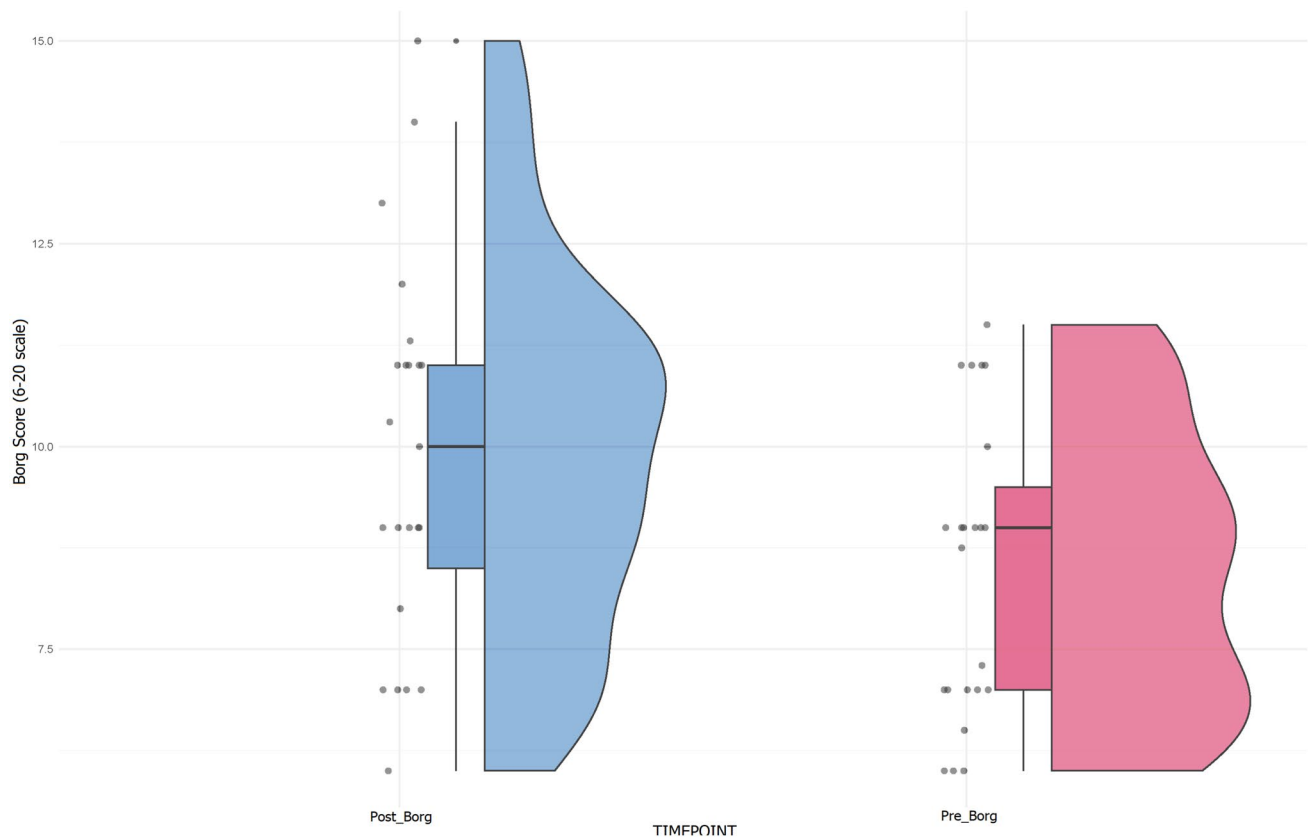


Fig. 6 Comparison of Fatigue Levels of Vessel Traffic Service Operators Before and After the Task

Table 5 Pre- and Post-Shift Borg RPE Effect Size Analysis

	Mean Borg RPE	SD	Median	Cohen's d
Pre Borg	8.48	1.82	9	0.68
Post Borg	9.90	2.35	10	0.68

significant differences (p -value = 0.031) between Post Borg and the shift (Table 5).

A comprehensive analysis of demographic factors revealed that age played a critical role in fatigue levels. Participants over 50 years of age exhibited the most significant variation in Borg RPE scores, indicating a higher degree of fatigue susceptibility compared to younger participants. Furthermore, shifts had a distinct impact, with night shift workers reporting the greatest variation in Borg scores, significantly more than morning or afternoon shifts. The influence of age, gender, and shift type on perceived fatigue levels is illustrated in Fig. 7 below, offering a clearer understanding of the interplay between these factors.

Although minimal, gender-related differences suggested slightly higher fatigue levels among male operators. Additionally, shifts had a distinct impact, with night shift workers

reporting the greatest variation in Borg scores, significantly more than morning or afternoon shifts.

The following Table 6 offers a synopsis of the mean Borg RPE scores for each shift type, along with the statistical significance of the results, both in the pre- and post-shift categories.

These results underscore the significant impact of night shifts on fatigue levels and highlight the necessity for targeted fatigue management strategies, particularly for older operators or those who regularly work night shifts.

5.1.1 Physiological measures: thermographic data

Thermographic data supported these findings, showing a nasal temperature decrease of approximately 2°C during high-demand tasks on night shifts, reflecting heightened fatigue levels. This analysis will be further explored in Section 5.2 to contextualize its implications.

5.2 Nasal temperature as a fatigue indicator

Thermographic imaging revealed a significant inverse correlation between nasal temperature and mental workload ($r = -0.78$, $p < 0.01$). These temperature changes highlight

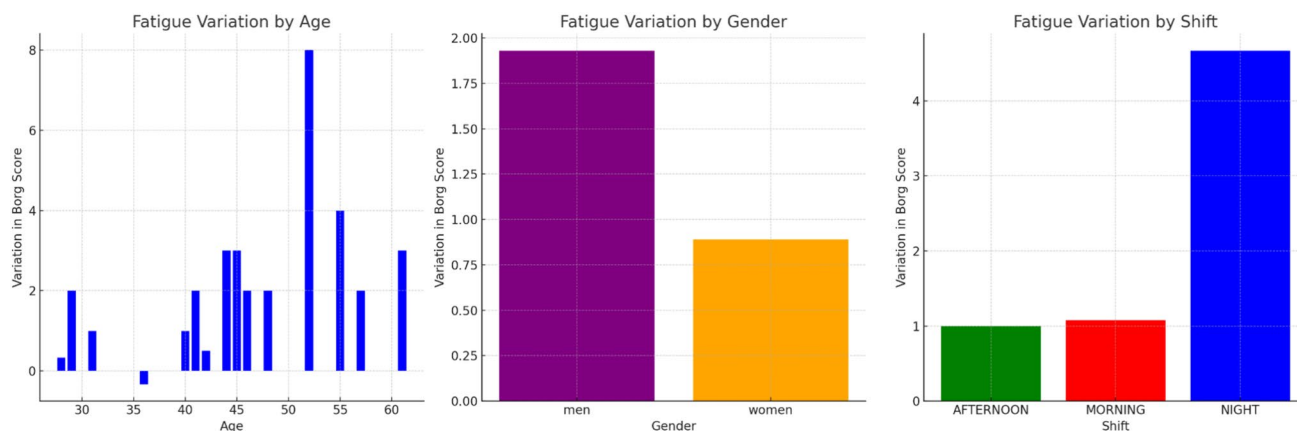


Fig. 7 Influence of Age, Gender, and Shift on VTSO Fatigue (Borg RPE)

Table 6 Borg RPE Scores Across Shift Types

Shift type	Mean Borg RPE (Pre)	Mean Borg RPE (Post)	Δ Borg (Post–Pre)	p-value (t-test)
Morning shift	10.2	11.3	+ 1.1	< 0.05
Afternoon shift	9.8	10.5	+ 0.7	< 0.05
Night shift	10.0	11.5	+ 1.5	< 0.01

the physiological response to increased mental workload and validate thermographic imaging as a reliable, real-time indicator of operator fatigue.

Since forehead temperature has not been related to workload in the literature, only the nasal temperature curve is shown to clearly illustrate this relationship. Fig. 8 shows nasal temperature fluctuations during the afternoon shift of a Vessel Traffic Services Operator (VTSO), with notable decreases observed during high demand tasks. Communication tasks are categorized as follows: Y = yacht, S =

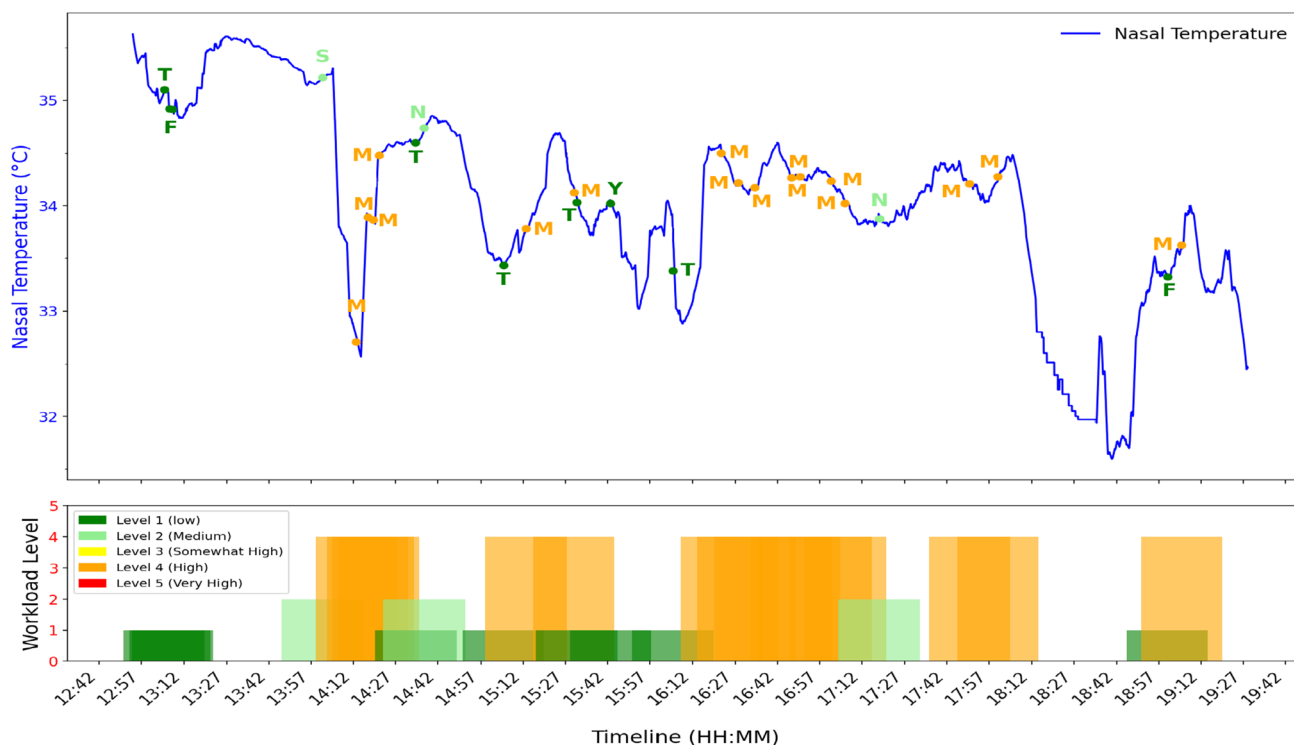


Fig. 8 Example of Workload-Related Nasal Temperature Variations During an Afternoon Shift of a VTSO (Letter and color codes represent VTSO tasks and cognitive demands, respectively, as shown in Table 7)

shipyard, F = fishing vessel, M = merchant vessel, N = naval vessel, and T = tour boat. The accompanying bar graph visually represents the Instantaneous Workload Scale (IWS): taller bars indicate higher workloads, with color codes indicating task categories and aligned with the timing of each event. A detailed analysis of nasal temperature trends in relation to workload is provided in the Results section.

As shown in the previous figure, nasal temperatures decreased rapidly as ship-to-shore communication increased, particularly for tasks involving merchant vessels or non-native languages. This trend was statistically significant during periods of high demand ($p < 0.01$), with temperature decreases corresponding to higher workload levels, particularly during merchant vessel-related tasks (orange) and emergency coordination tasks (red). The relationship between workload levels and nasal temperature is shown in Fig. 9. These results highlight the dynamic nature of thermographic monitoring, which provides a non-invasive and objective tool for assessing operator well-being in high-stress environments.

Fig. 9 Relationship Between Workload Levels and Nasal Temperature Variations

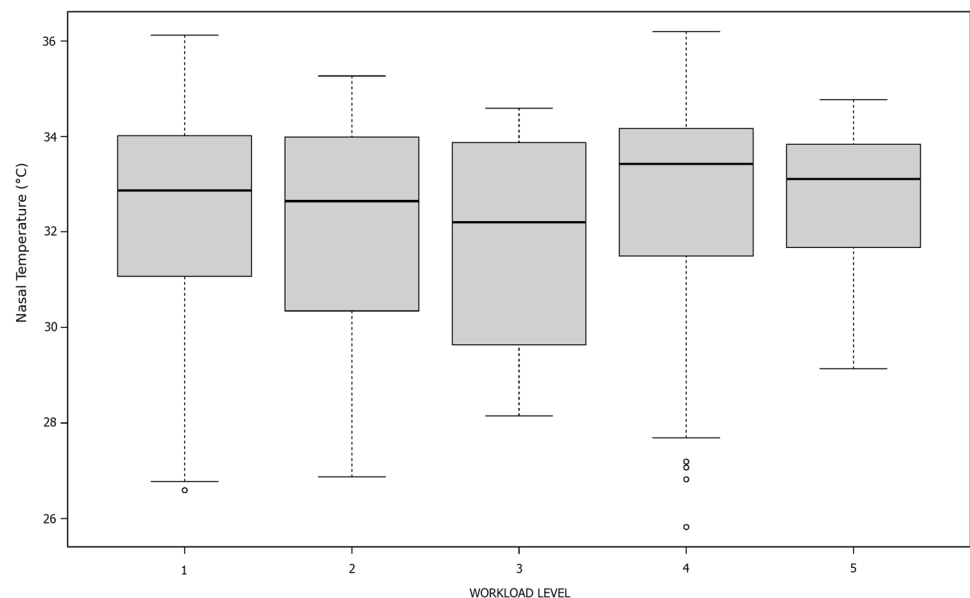


Table 7 Colorimetric and numeric scale of task performance in relation to workload

Workload level	Color indicator	Task/event category	Mental demand	Numerical value
Low	Green	Fishing Boats, Yacht Clubs, Barge Services, Cleaning Boats, Tourist Boats	Minimal	1
Medium	Light Green	Shipyards, Navy Ships	Moderate	2
Somewhat High	Yellow	Visitor presence	Elevated	3
High	Orange	Merchant ships	Significant	4
Very High	Red	Emergency coordination	Critical	5

5.3 Integrated workload scale (IWS) results

The Integrated Workload Scale (IWS) demonstrated its effectiveness in capturing workload variations across different shift types. Night shifts exhibited the highest workload scores (mean = 4.2, SD = 0.8), followed by morning (mean = 3.6, SD = 0.6) and afternoon shifts (mean = 3.2, SD = 0.5). Notably, emergency events caused a significant spike in workload metrics, with scores reaching critical levels (mean = 5.0, $p < 0.001$). These results underscore the heightened operational demands during night shifts and emergency scenarios.

Table 7 provides a detailed colorimetric and numeric scale, categorizing task performance by workload level and highlighting the mental demands associated with various operational events.

5.4 Relationships between rest and fatigue

Linear regression analysis revealed a strong inverse relationship between the duration of pre-task rest and subsequent fatigue levels ($p < 0.001$), indicating that participants

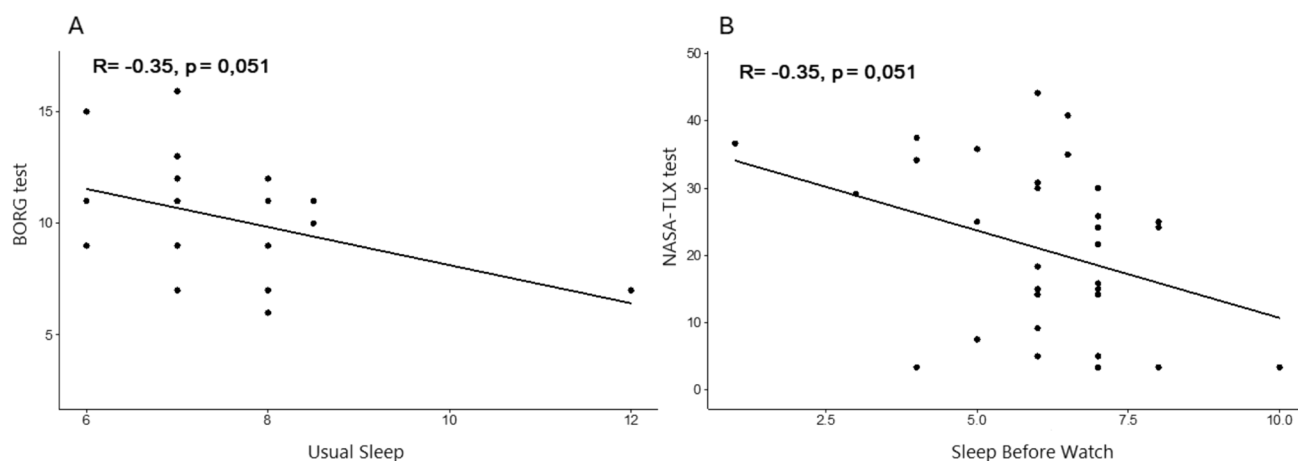


Fig. 10 Linear regression analysis showing the correlation between total usual sleep hours and Borg Scale scores (left) and the correlation between pre-task sleep hours and post-task NASA TLX scores (right)

who rested more prior to their shifts reported lower levels of perceived fatigue, as measured by the Borg Scale. Additionally, a negative correlation was observed between pre-task sleep duration and post-task NASA TLX scores emphasizing the influence of rest on perceived mental workload (Fig. 10).

To contextualize these findings, Fig. 11, presents the distribution of sleep duration reported by participants before their shifts. While this figure does not depict correlation, it visually highlights the variability in rest patterns among the sample. This variability is relevant in interpreting the strength of the relationships shown in previous figure, as it underscores how individual differences in sleep may contribute to variation in fatigue and workload perception.

Further analysis, shown in Fig. 12, examined the correlations between various fatigue and rest metrics, revealing several significant associations that offer insights into the relationship between sleepiness and fatigue. One of the key findings is the strong positive correlation between the Epworth Sleepiness Scale (ESS) and the SAM Valence scores, both pre- and post-watch. Specifically, the ESS was found to have a correlation of approximately 0.9 with SAM VALENCE PRE and 0.8 with SAM VALENCE POST, indicating that higher ESS scores are consistently associated with more negative mood states both before and after the watch period.

In addition, the SAM Arousal Post measure showed a strong correlation with SAM VALENCE POST ($r \approx 0.8$) and a moderate correlation with SAM VALENCE PRE ($r \approx$

Fig. 11 Distribution of Sleep Duration Before Shifts

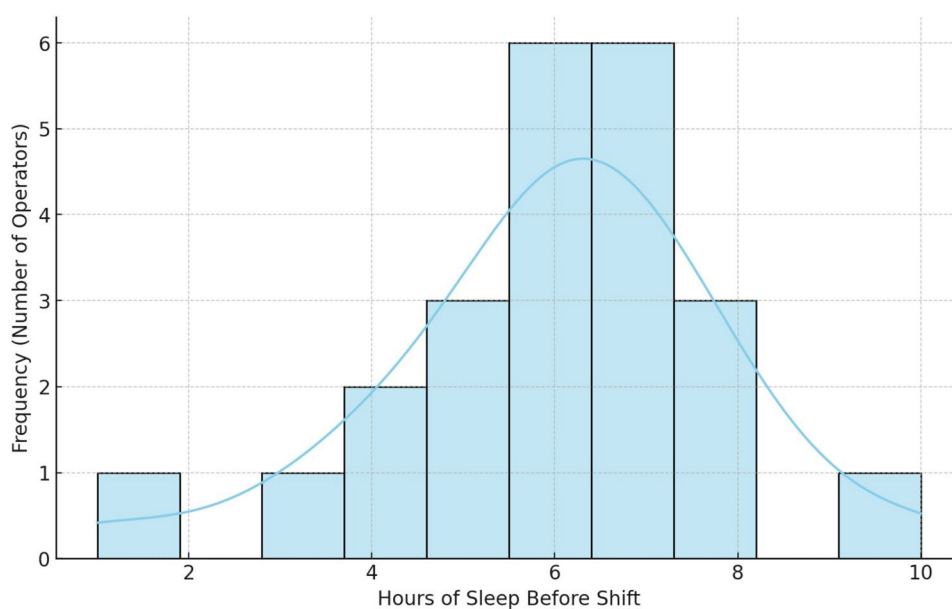
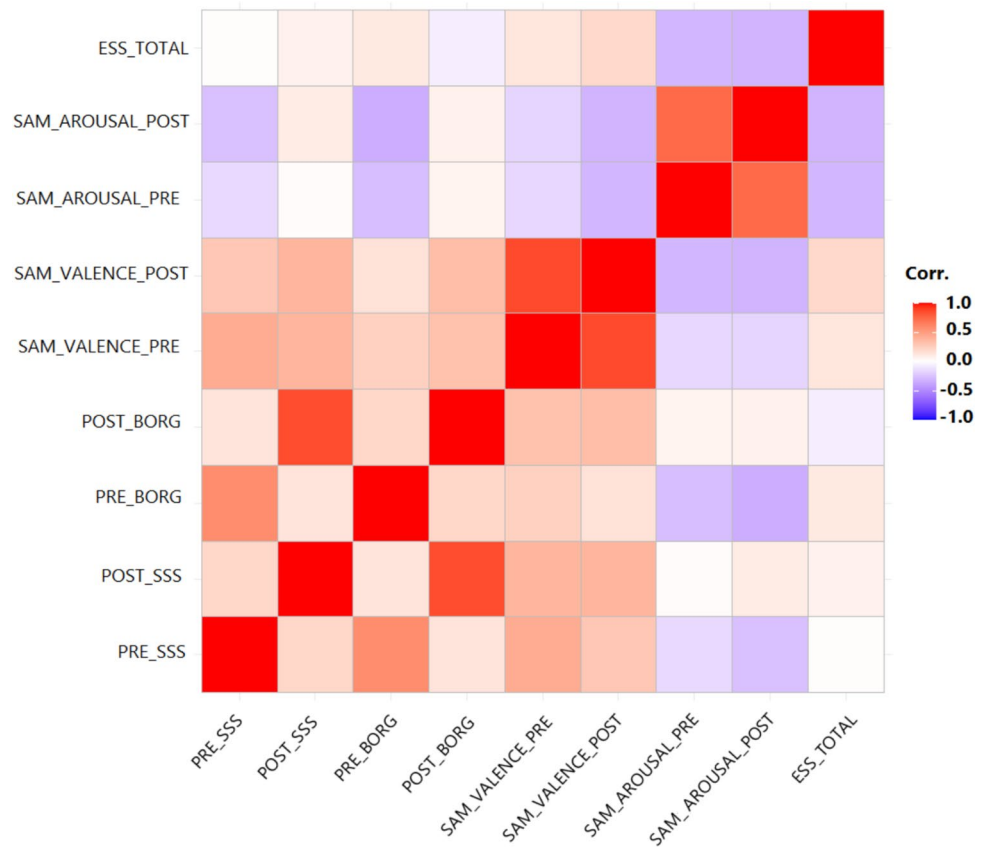


Fig. 12 Correlation matrix of various questionnaires assessing fatigue and rest used in the study (*SSS*=Stanford Sleepiness Scale; *ESS*=Epworth sleepiness scale; *SAM*=Self-Assessment Manikin *Borg*=Borg Rating of Perceived Exertion)



0.7), suggesting that increased arousal after the watch period is linked to both pre- and post-watch valence scores. Similarly, SAM Arousal Pre was found to be positively associated with SAM VALENCE PRE ($r \approx 0.9$) and moderately associated with SAM VALENCE POST ($r \approx 0.6$), suggesting that higher arousal before the watch period corresponds with more negative mood ratings both before and after the watch. Another notable finding is the very strong correlation between SAM VALENCE PRE and SAM VALENCE POST ($r \approx 0.9$), demonstrating a high degree of consistency in valence ratings across the pre- and post-watch periods. This consistency reinforces the idea that the overall mood state of the participants remains relatively stable before and after the watch.

Furthermore, moderate positive correlations were observed between the Pre and Post Borg scale ratings and fatigue measures such as Pre SSS ($r \approx 0.7$) and Post SSS ($r \approx 0.7$). This suggests a relationship between perceived fatigue, as measured by the Borg scale, and subjective sleepiness. Finally, the very strong correlation between Pre SSS and Post SSS ($r \approx 0.9$) suggests that participants' self-reported sleepiness levels remain relatively consistent over time.

These findings highlight the robustness and consistency of the correlations between the various fatigue and rest metrics, providing valuable insight into how sleepiness and fatigue interact and are measured across different scales.

5.5 Thermographic results

Thermographic analysis revealed significant fluctuations in nasal temperature that correlated with mental workload, particularly during high demand tasks such as coordinating with merchant vessels or managing long, monotonous periods. As a representative example, three operators from different shifts were analyzed to illustrate patterns observed across the full sample of 23 VTSOs.

Fig. 13a, shows data from a 36-year-old male operator collected during a seven-hour morning shift during which Thirty-six vessels were reported. All operators initially exhibited low nasal temperatures during the first hour, reflecting the cognitive demands of assuming the watch and establishing situational awareness. Two shaded areas indicate temperature drops approaching 4°C , coinciding with increased vessel calls to the VTS. This increase in workload corresponded to a decrease in nasal temperatures.

The first shaded interval, marked by high call density, showed erratic temperature patterns with frequent fluctuations. In contrast, the final hour, despite sustained call volume, displayed a rising temperature trend, likely due to operator self-activation in preparation for shift handover.

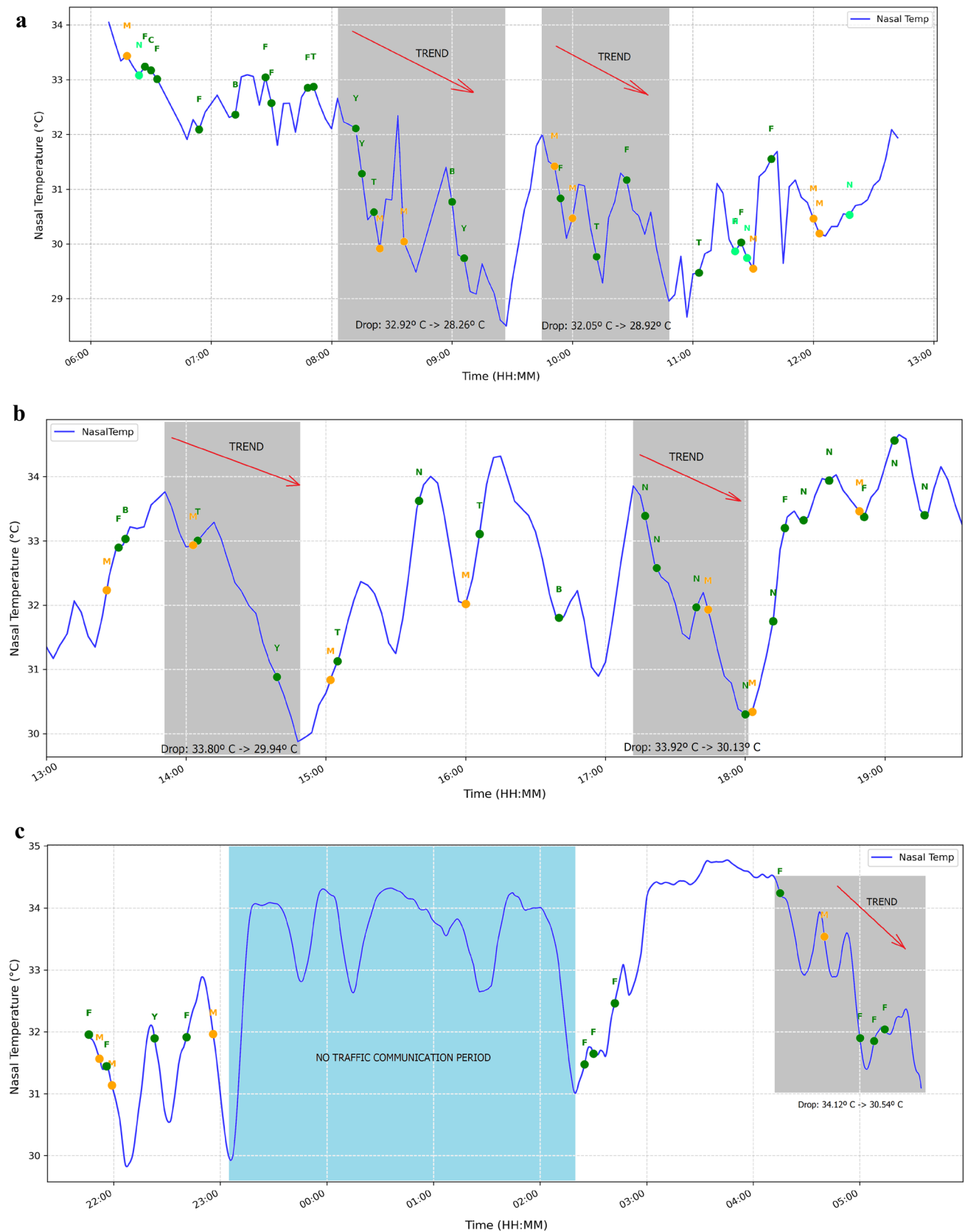


Fig. 13 a Effect of workload on nasal temperature during VTSO morning shift. b Effect of workload on nasal temperature during VTSO afternoon shift. c Effect of workload on nasal temperature during VTSO the night shift

The next analysis, shown in Fig. 13b, is for a 36-year-old female operator, collected during a seven-hour afternoon shift during which twenty-six vessels were reported. In this case, we observe the same trend, with two temperature drop patterns similar to the morning shift. These declines align with periods of increased operational demand, reinforcing the relationship between cognitive workload and reduced peripheral temperature. The repetition of this pattern across shifts suggests a consistent physiological response among operators to heightened task intensity. Notably, as in the morning shift, a gradual temperature increase is observed toward the end of the watch, likely reflecting anticipatory activation for the upcoming shift transition.

Morning and afternoon shifts showed greater fluctuations in nasal temperature, likely due to increased vessel traffic and mental workload (Table 8).

The final analysis, shown in Fig. 13c, relates to a 40-year-old male operator and was recorded during a ten-hour night shift during which fifteen vessels were reported. Compared to the day shift, the night shift, characterized by less traffic and less communication with vessels (indicated by the blue shaded area), showed more stable nasal temperature trends. Nevertheless, a notable decrease in temperature was observed towards the end of the shift, possibly reflecting circadian disruption and the cumulative effects of prolonged

task engagement. These observations are consistent with previous findings on physiological responses to mental workload (Or and Duffy 2007; Murai et al. 2015; Abdelrahman et al. 2017; Cho et al. 2019).

Temperature fluctuations align with the Integrated Workload Scale (IWS) results, supporting nasal temperature as a reliable, real-time indicator of operator fatigue. The afternoon shift exhibited a temperature range of 30°C to 36°C, with a median of 34°C, indicating moderate variability in fatigue. The morning shift showed the broadest range (28°C to 36°C), reflecting consistent operational demands. In contrast, the night shift displayed a more stable range around 32°C, suggesting a higher, sustained level of fatigue (Fig. 14). ANOVA analyses revealed significant relationships between nasal temperature and gender ($p = 2e-16$) and workload level ($p = 0.00927$), confirming the observed variability.

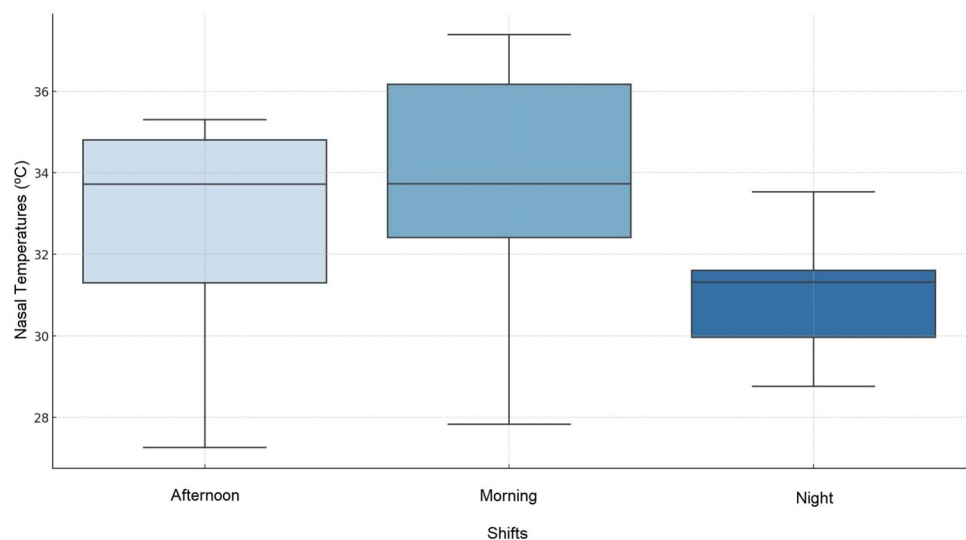
6 Discussion

This study provides solid evidence that nasal temperature, as measured by facial thermography, serves as a reliable, real-time physiological indicator of mental workload and fatigue in vessel traffic service operators (VTSOs). The observed inverse correlation between nasal temperature and workload intensity is consistent with previous research in high-stress domains such as air traffic control and automotive systems, where cognitive demands have been associated with peripheral vasoconstriction and subsequent temperature decreases (Or and Duffy 2007; Murai et al. 2015; Marinescu et al. 2018; Cho et al. 2019; Diaz-Piedra et al. 2019). This physiological trend, particularly the sharp decline in nasal temperature during high-demand periods and its stabilization or recovery during lower-demand phases, suggests a consistent autonomic response to cognitive strain.

Table 8 Average Number of Vessels Reported by Shift

Shifts	Number of shifts recordings	Average number of vessels Reported
Morning	10	21
Afternoon	16	25.8
Evening	6	14.5

Fig. 14 VTSO's Nasal Temperature Variability per Shift



By integrating thermographic data with validated subjective instruments (Borg RPE, SSS, NASA-TLX), and introducing the domain-specific Integrated Workload Scale (IWS), the study applies a comprehensive and multidimensional approach to fatigue assessment. The correlation between IWS workload levels and nasal temperature fluctuations provides empirical support for the scale's sensitivity and applicability in real operational settings. This triangulation strengthens the internal validity of the findings and provides a nuanced perspective on how cognitive demands manifest physiologically throughout a shift.

A key advantage of the methodology lies in its non-intrusiveness. Unlike EEG or wearable biosensors, which may disrupt natural workflows or require physical contact, thermographic imaging allows continuous monitoring without interfering with the operator's task performance. Furthermore, the implementation of the YOLO5Face algorithm for automatic landmark detection significantly increased the speed and reliability of data processing, offering a scalable solution for real-world fatigue monitoring.

However, limitations must be acknowledged. First, while thermographic imaging proved effective, the accuracy of measurements can be influenced by environmental factors (e.g., room temperature, humidity) and inter-individual variability such as age, skin emissivity, and health status. Despite rigorous standardization, some residual uncertainty in thermal readings is inevitable. In addition, the Thermal Face in the Wild (TFW) dataset used to train the detection algorithm may lack adequate demographic diversity, which could impact generalizability across populations.

Furthermore, nasal temperature patterns observed throughout the shifts offer important physiological insights. During high workload tasks, especially those involving emergency management or coordination with merchant vessels, nasal temperature exhibited rapid declines, often as much as 2–4°C. Toward the end of shifts, however, temperature values tended to rise slightly, potentially reflecting anticipatory arousal or cognitive adjustment during the transition to task completion. In contrast, night shifts showed lower but more stable nasal temperature baselines, likely linked to circadian fatigue and lower environmental stimulation, rather than acute workload peaks. These findings are consistent with existing research in safety-critical industries such as aviation and air traffic control, where shift work and workload are similarly associated with fatigue and operational risk (Gawron 2008; Caldwell et al. 2019).

Collectively, these findings make a significant contribution to the literature by demonstrating how thermal imaging can be integrated into socio-technical systems to assess operator well-being in safety-critical environments. They also underscore the importance of shift design, particularly in mitigating the physiological toll of night shifts and supporting older operators who are more susceptible to fatigue.

Future studies will attempt to replicate the research in other MRCCs in order to increase the sample size, to corroborate the findings, and to minimize the limitations that were identified.

7 Conclusions

This study confirms that fatigue is a critical maritime safety issue that significantly impacts the performance and well-being of VTSOs. By integrating thermographic imaging with traditional subjective measures, it demonstrates a dual approach that provides a more comprehensive understanding of fatigue dynamics. Thermographic imaging, in particular, offers objective, real-time data that aligns with subjective reports, enabling targeted and effective fatigue management strategies. The findings highlight the substantial impact of shift timing, particularly night shifts, on fatigue levels due to circadian misalignment and cognitive demands during high-traffic periods.

Key recommendations to mitigate fatigue include optimizing shift scheduling by minimizing consecutive night shifts and aligning rotations with circadian rhythms. Real-time monitoring through thermographic imaging can help identify periods of high fatigue, allowing for timely intervention. In addition, age-sensitive policies should be developed that offer strategies for older operators, such as extended recovery periods and personalized wellness programs. To support cognitive performance, the introduction of micro-breaks and cognitive support tools during peak workload periods is essential. Finally, promoting sleep hygiene by encouraging pre-shift rest and optimizing sleep habits will contribute to overall operator well-being and performance.

Future research should explore the integration of thermographic tools into safety management systems, their cost-effectiveness, and their long-term application in other safety-critical industries. In addition, the evaluation of tailored fatigue management programs can further improve the safety and performance of operators worldwide.

In conclusion, addressing fatigue through a combination of innovative monitoring, shift optimization, and tailored support strategies will not only improve the safety and performance of VTSOs, but also contribute to a broader culture of safety in industries where human performance is critical.

In conclusion, this study demonstrates the feasibility and effectiveness of thermographic imaging for fatigue monitoring in maritime contexts. While providing a novel methodological approach, it also lays the groundwork for broader implementation in high-risk industries. Future research should prioritize cross-center replication, address demographic variability, and further refine automated thermal detection tools to improve their adaptability and accuracy in diverse operational environments.

Appendix A

Study demographic sheets and questionnaires

The following are the data sheets and questionnaires used for this study.

Data sheet

Date of Interview	dd	mm	yy
Task Start time	hh	mm	ss
Task End time	hh	mm	ss
Date of Birth	dd	mm	yy
Sex	☐ Male	☐ Female	☐ no binary
Dominant Hand	☐ Left ☐ Right		

ID Number		
Age		years
Height		cm
Weight		Kg

Drinks Consumption			Quantity		Type of beverage (Wine, beer, coffee, coke...)
Do you drink alcohol?	☐ Yes	☐ No	No. of glasses		
Do you drink caffeinated beverages?	☐ Yes	☐ No	Can or cup count		

(a standard drink is considered 12 ounces of beer, 5 ounces of wine, or 1.5 ounces of distilled spirits)

How many hours do you usually sleep?		Hours
How many hours did you sleep last night?		Hours
How long has it been since you woke up this morning?		Hours
Do you have night shifts?	☐ Yes	☐ No

Other observations

Years of experience in your current MRCC	☐ Gijón ☐ Cartagena	_____ years
Total VTSO experience	_____ years	
Specialty	☐ Supervisor ☐ VTSO	

Total VTSO seniority (years)	<2	2 a 5	5 a 10	10 a 20	>20
	☐ Junior	☐ Middle	☐ Senior	☐ Fellow	☐ Expert

Marital Status	☐ Single		☐ Married/ Stable partner		☐ Widow		☐ Divorced	
Children	☐ Yes	☐ No	How many?		Ages			
Education level	☐ No studies		☐ Primary		☐ Secondary		☐ VT	☐ University

Do you have any diagnosed disease/disorder?	☐ Yes	☐ No
☐ Hypertension		
☐ Diabetes		
☐ High cholesterol		
☐ Cardiovascular diseases		
☐ Pulmonary diseases/Allergies		
☐ Endocrine diseases		
☐ Neurological diseases		
☐ Gastrointestinal diseases		
☐ Rheumatic diseases		
☐ Psychopathological disorder	☐ Anxiety ☐ Depression ☐ Others	
☐ Other diseases		

Do you take any medicines regularly?		☐ Yes	☐ No
Commercial Name	Dose	Daily Frequency	Start Date
Have you taken any medicines in the last few hours?		☐ Yes	☐ No

If yes, specify type and quantity

Trade name		Quantity	
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Scales

Morningness-eveningness questionnaire (MEQ)

The following questions relate to your sleep/wake cycle. Please try to answer as accurately as possible by thinking about your daily life. Read each question carefully before answering. Answer ALL the questions. Answer in numerical order. Each question must be answered independently of the previous one. Do not go back to check your answers.

All questions have more than one possible answer. Please tick only one option for each question. Mark with an 'x' next to the answer that best describes you.

What time would you get up if you were entirely free to plan your day?												
AM	6	7	8	9	10	11	12	1	PM			
During the first half hour after you wake up in the morning, how do you feel?												
☒ Very tired					☐ Fairly refreshed							
☒ Fairly tired					☐ Very refreshed							
At about what time in the evening do you feel tired and need to sleep?												
PM	9	10	11	12	1	2	3	4	AM			
At approximately what time of day do you usually feel your best?												
12	1	2	3	4	5	6	7	8	9	10	11	12
Midnight											Midday	Midnight
One hears about "morning types" and "evening types." Which one of these types do you consider yourself to be?												
☐ Definitely a morning type						☐ Rather more an evening type						
☐ Rather more a morning type						☐ Rather more an evening type						
☐ Definitely an evening type						☐ Rather more a morning type						

Epworth sleepiness scale

The following scale lists eight situations associated with different levels of sleepiness. Using the scale below, indicate how likely you are to fall asleep in each situation.

Indicate the answer that best describes you in a normal situation.

0 = No chance of dozing

1 = Slight chance of dozing

2 = Moderate chance of dozing

3 = High chance of dozing

Situation	Chance of Dozing
Sitting and reading	
Watching TV	
Sitting inactive in a public place (e.g., a theatre or a meeting)	
As a passenger in a car for an hour without a break	
Lying down to rest in the afternoon when circumstances permit	
Sitting and talking to someone	
Sitting quietly after a lunch without alcohol	
In a car, while stopped for a few minutes in traffic	

Stanford sleepiness scale (SSS)

Put a cross in the box corresponding to the state with which you best identify AT THIS TIME.

1. Feeling active and vital, alert, wide awake: You feel fully alert and awake, with no feeling of sleepiness or fatigue.
2. Relaxed, awake, not feeling sleepy: You feel relaxed and awake, but not sleepy or tired.
3. Neither sleepy nor not sleepy: You feel neither sleepy nor not sleepy, but rather neutral.

4. Fighting sleep, but not asleep: You are trying to stay awake, but feel a strong desire to sleep.
5. Sleepy, but not asleep: You are sleepy, but not yet asleep.
6. Sleepy, but dozing off: You are sleepy and dozing off, but still somewhat awake.
7. Almost in reverie, sleep onset soon, lost struggle to remain awake: You are almost asleep, with a strong urge to sleep and a struggle to stay awake.

PRE self-assessment-manikin scale (SAM)

The first scale (first row of dolls) corresponds to the valence dimension, ranging from a smiling doll to a serious doll. If you now feel happy, satisfied, pleased, content, mark the dummy on the left with an X. If you now feel unhappy, annoyed or dissatisfied, mark the dummy on the right. If you

(PRE)	1st hour	2nd hour	3er hour	4th hour	5th hour	6th hour	7th hour	8th hour	9th hour	10th hour	(POST)

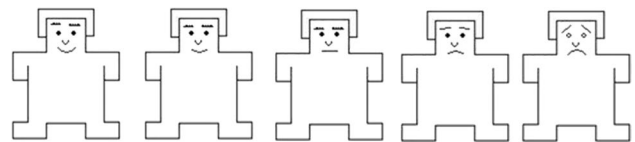
BORG rating of perceived exertion scale

This numerical scale ranges from 6 to 20. Try to assess your sense of fatigue as honestly as possible. Do not underestimate, but do not overestimate either. It is your sense of fatigue and exertion that is important, not how it compares to that of others. 6 means “no perceived fatigue” and 20 means “maximum perceived fatigue”.

(PRE)	1st hour	2nd hour	3er hour	4th hour	5th hour	6th hour	7th hour	8th hour	9th hour	10th hour	(POST)
Show 2.7											show 2.9

now feel unhappy, annoyed or dissatisfied, put an X on the doll on the right. You can also express intermediate feelings of pleasure. If you feel completely neutral, i.e., neither happy nor sad, mark with an X on one of the middle dummies. If your feelings of pleasure or dissatisfaction fall between two dolls, put an X in the rectangle between them.

Rate how positive or negative the emotion is that you feel, ranging from unpleasant feelings to pleasant feelings of happiness



NASA-TLX (Task Load Index)

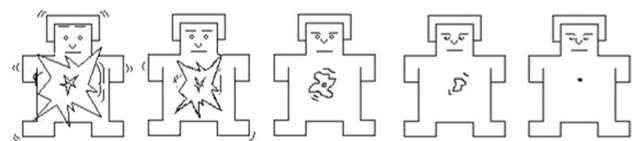
The NASA-TLX (Task Load Index) is an instrument based on the assumption that mental workload is a hypothetical construct that represents the cost incurred by the worker in trying to achieve a specific level of performance. This numerical scale ranges from 0 to 100. Try to assess your sense of mental workload as honestly as possible. Do not underestimate, but do not overestimate either. It is your sense of mental workload that is important, not how it compares to that of others.

0 means “no perceived mental workload” and 100 means “maximum perceived mental workload”

(PRE)	1st hour	2nd hour	3er hour	4th hour	5th hour	6th hour	7th hour	8th hour	9th hour	10th hour	(POST)
N/A											Show 3.0

The second scale (second row of dummies) corresponds to the arousal dimension or activation and goes from an excited dummy to a calm dummy. If you now feel stimulated, excited, agitated or active, put an X on the left dummy. If you now feel completely relaxed, calm, inactive, mark with an X the rightmost dummy. As in the previous dimension, you can also place intermediate levels of excitement or calm.

Rate how excited or apathetic the emotion is that you feel, ranging from frantic excitement to sleepiness or boredom.



PRE BORG rating of perceived exertion scale

INSTRUCTIONS:

This numerical scale, ranges from 6 to 20.

6 means "no perceived fatigue" and 20 means "maximum perceived fatigue".

Try to assess your sense of fatigue as honestly as possible. Do not underestimate, but do not overestimate either. It is your sense of fatigue and exertion that is important, not how it compares to that of others.

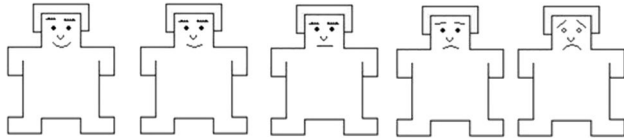
Score	Level of exertion	
6.	No exertion at all	☐
7.	Extremely light	☐
8.		☐
9.	Very light	☐
10.		☐
11.	Light	☐
12.		☐
13.	Somewhat hard	☐
14.		☐
15.	Hard (heavy)	☐
16.		☐
17.	Very hard	☐
18.		☐
19.	Extremely hard	☐
20.	Maximal exertion	☐

POST self-assessment-manikin scale (SAM)

The first scale (first row of dolls) corresponds to the valence dimension, ranging from a smiling doll to a serious doll. If you now feel happy, satisfied, pleased, content, mark the dummy on the left with an X. If you now feel unhappy, annoyed or dissatisfied, mark the dummy on the right. If you now feel unhappy, annoyed or dissatisfied, put an X on the doll on the right. You can also express intermediate feelings

of pleasure. If you feel completely neutral, i.e., neither happy nor sad, mark with an X on one of the middle dummies. If your feelings of pleasure or dissatisfaction fall between two dolls, put an X in the rectangle between them.

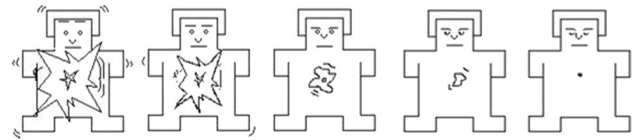
Rate how positive or negative the emotion is that you feel, ranging from unpleasant feelings to pleasant feelings of happiness



The second scale (second row of dummies) corresponds to the arousal dimension or activation and goes from an excited dummy to a calm dummy. If you now feel stimulated, excited, agitated or active, put an X on the left dummy. If you now feel completely relaxed, calm, inactive, mark with

an X the rightmost dummy. As in the previous dimension, you can also place intermediate levels of excitement or calm.

Rate how excited or apathetic the emotion is that you feel, ranging from frantic excitement to sleepiness or boredom.



POST BORG rating of perceived exertion scale**INSTRUCTIONS:**

This numerical scale, ranges from 6 to 20.

6 means "no perceived fatigue" and 20 means "maximum perceived fatigue".

Try to assess your sense of fatigue as honestly as possible. Do not underestimate, but do not overestimate either. It is your sense of fatigue and exertion that is important, not how it compares to that of others.

Score	Level of exertion	
6.	No exertion at all	☐
7.	Extremely light	☐
8.		☐
9.	Very light	☐
10.		☐
11.	Light	☐
12.		☐
13.	Somewhat hard	☐
14.		☐
15.	Hard (heavy)	☐
16.		☐
17.	Very hard	☐
18.		☐
19.	Extremely hard	☐
20.	Maximal exertion	☐

NASA—task load index (POST)

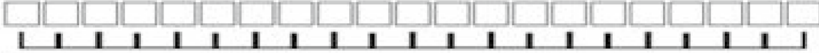
The NASA-TLX (Task Load Index) is an instrument based on the assumption that mental workload is a hypothetical construct that represents the cost incurred by the worker in trying to achieve a specific level of performance. Try to rate your performance across the following six dimensions

Mental demand. The amount of mental and perceptual activity required by the task (e.g. thinking, deciding, calculating, remembering, looking, searching, etc.).

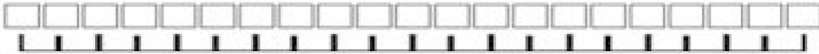
How mentally demanding was the task?

Very Low  Very High

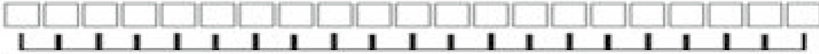
Physical demand. The amount of physical activity required by the task (e.g., pushing, pulling, turning, etc.). How physically demanding was the task?

Very Low  Very High

Temporal demand. Level of time pressure felt. Ratio of time required to time available. How hurried or rushed was the pace of the task?

Very Low  Very High

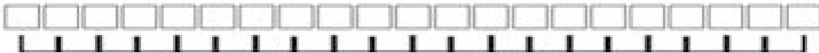
Effort. The degree of mental and physical effort expended to achieve an adequate level of performance. How hard did you have to work to accomplish your level of performance?

Very Low  Very High

Performance. The extent to which the individual is satisfied with his or her level of performance. How successful were you in accomplishing what you were asked to do?

Perfect  Failure

Frustration. The extent to which the subject feels insecure, stressed, irritated, dissatisfied, etc. during the performance of the task. How insecure, discouraged, irritated, stressed, and annoyed were you?

Very Low  Very High

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Author contributions F.C.M. conceptualized the study, designed the methodology, conducted the investigation, curated the data, and drafted the original manuscript. J.R.G. contributed to the methodology, investigation, data curation, and manuscript writing. J.S.M. participated in the study's methodology, investigation, manuscript writing, and provided supervision throughout the research process. R.L.A. was involved in the methodology, investigation, data curation, and manuscript writing. All authors reviewed, revised, and approved the final version of the manuscript and agree to be accountable for its content.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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